R = Resistance (in ohms)

X_C = Inductive capitance.

$V_R = I \times R$

$V_C = I \times X_C$

$V^2 = V_R^2 + V_C^2$

$V = \sqrt{V_R^2 + V_C^2}$

$V = \sqrt{(I_R)^2 + (I\,X_C)^2}$

$V = \sqrt{I^2R^2 + I^2\,X_C^2}$

$V = I\ \sqrt{R^2 + X_C^2}$

$\frac{V}{I} = Z = \sqrt{R^2 + X_C^2}$

5.4.8 IMPEDANCE

Impedance is the total opposition offered by the circuits elements [ie, Resistance, Inductance and capacitance] simply, Impedance is defined as the ratio of the voltage to current

$$\text{Impedance}(Z) = \frac{\text{Voltage}(V)}{\text{Current}(I)}$$

5.4.9 RLC Series Circuit

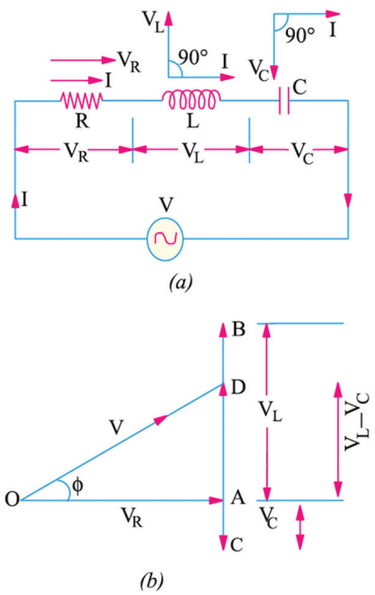


Fig 5.12 *AC through RLC Series Circuit*

In this RLC circuit resistance, inductance and capacitance are connected in series. In this the current is same. The voltage is differed by circuit elements, Total supply voltage is 'V'

$I = I_R = I_L = Ic$

And

$V = V_R + V_L + V_C$

V_R = Voltage across the resistance

V_L = Voltage across the inductance

V_C = Voltage across the capacitance

And also

V_R is in phase with current

V_L leads current by 90°

V_C lags current by 90°

(i) If inductive reactance is greater than capacitive reactance ($X_L > X_C$)

$V^2 = V_R^2 + (V_L - V_C)^2$

$V = \sqrt{V_R^2 + (V_L - V_C)^2}$

$V = \sqrt{(IR)^2 + (IX_L - IX_C)^2}$

$V = \sqrt{I^2R^2 + I^2(X_L - X_C)^2}$

$V = I\sqrt{R^2 + (X_L - X_C)^2}$

$\frac{V}{I} = \sqrt{R^2 + (X_L - X_C)^2}$

$$\text{Impedance } Z = \sqrt{R^2 + (X_L > X_c)}\ 2 \text{ ohms}$$

$$\left(\frac{V}{I} = Z\right)$$

$$\text{Power factor } \cos\theta = \frac{R}{Z}$$

and power $P = VI\ \cos\theta$ watts

(ii) If capacitive reactance is greater than Inductive reactance ($X_C > X_L$)

$$\text{Impedance } Z = \sqrt{R^2 + (X_C > X_L)^2} \text{ ohms}$$

$$\text{Power factor } \cos\theta = \frac{R}{Z}$$

And power $P = VI\ \cos\theta$ watts

EMA_English_Chp05.indd 79 11/9/2020 3:09:12 PM